

# Natural variation in salt-induced changes in root:shoot ratio reveals *SR3G* as a negative regulator of root suberization and salt resilience in *Arabidopsis*

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## Abstract

Soil salinity is one of the major threats to agricultural productivity worldwide. Salt stress exposure alters root and shoot growth rates, thereby affecting overall plant performance. While past studies have extensively documented the effect of salt stress on root elongation and shoot development separately, here we take an innovative approach by examining the coordination of root and shoot growth under salt stress conditions. Utilizing a newly developed tool for quantifying the root:shoot ratio in agar-grown *Arabidopsis* seedlings, we found that salt stress results in a loss of coordination between root and shoot growth rates. We identify a specific gene cluster encoding domain-of-unknown-function 247 (DUF247), and characterize one of these genes as Salt Root:shoot Ratio Regulator Gene (*SR3G*). Further analysis elucidates the role of *SR3G* as a negative regulator of salt stress tolerance, revealing its function in regulating shoot growth, root suberization, and sodium accumulation. We further characterize that *SR3G* expression is modulated by *WRKY75* transcription factor, known as a positive regulator of salt stress tolerance. Finally, we show that the salt stress sensitivity of *wrky75* mutant is completely diminished when it is combined with *sr3g* mutation. Together, our results demonstrate that utilizing root:shoot ratio as an architectural feature leads to the discovery of new stress resilience gene. The study's innovative approach and findings not only contribute to our understanding of plant stress tolerance mechanisms but also open new avenues for genetic and agronomic strategies to enhance crop environmental resilience.

### eLife assessment

Through cellular, developmental, and physiological analysis, this **valuable** study identifies a gene that functions to regulate the relative growth of roots and shoots under salt stress. The holistic approach taken provides **solid** evidence that this gene, a member of a larger tandemly duplicated gene family initially highlighted by association mapping, as well as an upstream regulator contribute to salt tolerance. More robust statistical or biological support for some conclusions could further strengthen this manuscript. The manuscript will be of interest to plant biologists studying mechanisms of abiotic stress tolerance and gene family evolution.

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## Introduction

Salt stress is predominant in the dryland areas where evaporation rate exceeds water input, resulting in increased accumulation of salts in the upper soil layers (Abrol, Yadav, and Massoud 1988; Fischer et al. 2021; Richards 1969). Dryland areas of South America, southern and western Australia, Mexico, southwest United States, and South Africa were previously identified as primary hotspots for soil salinization (Hassani, Azapagic, and Shokri 2021). As fresh water becomes scarcer, understanding the mechanisms of salt stress resilience becomes of paramount importance in the context of global agricultural productivity and sustainability.

Salt stress detrimentally affects plant growth and development, decreasing the activity of the meristem (West, Inzé, and Beemster 2004). Studies utilizing main root elongation as an indicator of salt tolerance have led to identification of many crucial components of salt stress signaling and tolerance, including genes involved in salt-overly sensitive signaling (SOS) pathway, jasmonate, ethylene, abscisic acid (ABA), reactive oxygen species (ROS) in glycophytes and halophytes alike (Wu, Ding, and Zhu 1996; Valenzuela et al. 2016; Yunfei Hu et al. 2023; Geng et al. 2013; Fu et al. 2019; Şekerçi et al. 2023). Additionally, negative halotropism response, that describes a directional growth of main root away from the salt gradient, was used to identify a wide range of molecular components regulating the auxin redistribution and cell-wall remodeling, to actively avoid high salt (Galvan-Ampudia et al. 2013; Deolu-Ajayi et al. 2019; B. Yu et al. 2022; Korver et al. 2020; Y. Zou et al. 2022). The development of lateral roots is affected by salinity, and the differential effect on main and lateral roots within specific genotype can result in substantial reprogramming of root system architecture (Julkowska et al. 2014, 2017). The shoot development is impacted by salt stress, as the salt impacts the plant transpiration and photosynthetic efficiency instantaneously (Awlia et al. 2016, 2021; Julkowska et al. 2016). Many forward genetic studies were highly successful in associating natural variation in root and shoot growth with underlying genes (Awlia et al. 2021; Julkowska et al. 2016, 2017). However, these studies typically focus on either above or below ground tissue.

The coordination of root and shoot growth rate is an important aspect of seedling establishment, and becomes even more critical under abiotic stress (Karcher et al. 2008; Xu et al. 2015; Chen et al. 2022; Y. Wang et al. 2018; Bacher et al. 2022). The importance of increased root:shoot ratio, maximizing water scavenging while limiting transpiration, is well studied under drought stress (Asch et al. 2005; Karcher et al. 2008; Uga et al. 2013; Chen et al. 2022). However, the effect of salt stress on root:shoot ratio is yet to be described, and its contribution to salt stress remains unknown. Here, we developed a tool to quantify root:shoot ratio from agar-grown *Arabidopsis* seedlings, and re-analyzed the natural diversity panel of *Arabidopsis* exposed to salt

stress (Julkowska et al. 2017 [DOI](#)). We have observed that while salt stress does not elicit a clear response in root:shoot ratio, it results in loss of coordination of root and shoot growth rates. The locus associated with salt-induced changes in root:shoot ratio consisted of a gene cluster encoding domain-of-unknown-function 247 (DUF247), with majority of SNPs clustering into one specific gene (At3g50160), that we named as Salt Root:shoot Ratio Regulator Gene (SR3G). Further characterization of SR3G revealed its function as a negative regulator of salt stress resilience, through regulation of root suberization, shoot growth, and  $\text{Na}^+$  accumulation. We found WRKY75 transcription factor, a positive regulator of salt stress tolerance, acts upstream of SR3G, binding directly to the SR3G promoter in yeast. While *sr3g* and *wrky75* mutants showed contrasting salt stress response, the double mutants of *sr3g/wrky75* displayed reduced sensitivity to salt stress.

## Materials & Methods

### Screening HapMap population for the increase in root and shoot size

The images collected within (Julkowska et al. 2017 [DOI](#)) were re-analyzed in this manuscript to examine salt-induced changes in the root:shoot ratio for the data presented here. 360 accessions of the Arabidopsis HapMap population (Weigel and Mott 2009 [DOI](#)) were screened over 7 experimental batches, using 4 biological replicates per genotype per condition. The seeds were surface sterilized, incubated in sterile MQ water at 4°C for 48 h, and germinated on media containing ½ Murashige-Skoog, 0.5% sucrose, 0.1% MES buffer, pH 5.8 (KOH) and 1% Dashin agar. The 4-day-old seedlings were subsequently transferred to media additionally supplemented with 0, 75 or 125 mM NaCl. The plates were scanned every 2nd day using a flat-bed scanner for 8 consecutive days.

The collected images were analyzed for green and white pixels, corresponding to the projected surface of the shoot and root, respectively (<https://github.com/ronellsicat/PlantSeg> [DOI](#)). A Matlab-based tool was developed and used to simplify and speed up the segmentation and analysis pipeline. For automatic segmentation, the tool uses a combination of image operations (histogram equalization), thresholding on different color spaces (e.g., RGB, YCbCr, Lab, HSV), and binary image processing (boundary and islands removal). For analysis, the tool uses connected components analysis to extract the plants (combined roots and shoots) and can guide the user to find possible mistakes which can be manually corrected via the user interface. After segmentation, the number of pixels for roots and shoots are counted and recorded automatically. More details are provided in the publicly available repository for the tool (<https://github.com/ronellsicat/PlantSeg> [DOI](#)). The collected images and corresponding data acquired from them using the tool can be accessed at Zenodo Repository (experimental batch 1: <https://doi.org/10.5281/zenodo.7272155> [DOI](#); experimental batch 2: <https://doi.org/10.5281/zenodo.7272344> [DOI](#); experimental batch 3: <https://doi.org/10.5281/zenodo.7272647> [DOI](#); experimental batch 4: <https://doi.org/10.5281/zenodo.7271841> [DOI](#); experimental batch 5: <https://doi.org/10.5281/zenodo.7272139> [DOI](#); experimental batch 6: <https://doi.org/10.5281/zenodo.7273936> [DOI](#); experimental batch 7: <https://doi.org/10.5281/zenodo.7273937> [DOI](#); quantified data: <https://doi.org/10.5281/zenodo.7268673> [DOI](#)). The tool's precision was evaluated by growing the Col-0 seedlings on agar plates, analyzing their root and shoot projected area on the final day using the tool, and correlating it with the recorded fresh and dry weight of the seedlings (Fig. S1). The collected data from the HapMap panel was analyzed using R for potential outliers (samples identified to be more than 3x standard deviation from the mean of each genotype within each condition), and the genotype-specific mean was calculated using the stats package (Team et al. 2018 [DOI](#)). The overall increase in root and shoot area was investigated for differences between control and salt stress conditions (Fig. S2). The root and shoot growth were estimated by fitting a linear function for a log-transformed projected root or shoot area over the non-transformed timeline of the experiment (<https://rpubs.com/mjulkowska/HapMap-root-shoot-data> [DOI](#)). The data were subsequently used to explore differences between the experimental batches and the standard deviations within individual accessions. The genotypes above 3 standard deviations from

the population's mean were excluded from further analysis. The salt tolerance index (STI) was calculated for root and shoot growth by dividing the average trait value at salt stress by the genotype-specific average at control conditions (**Fig. S3**). The correlations between individual traits were examined using the genotypic mean (**Fig. S4**). All quantitative data for root and shoot growth was visualized using ggplot2, corrplot and ggpubr packages (Kassambara, n.d. [\[1\]](#); Wilke 2016 [\[2\]](#); Wei and Simko 2017 [\[3\]](#)). For all measured traits, we estimated broad sense heritability ( $H^2$ ) using the MVApp (Julkowska et al. 2018 [\[4\]](#)) (**Table S1**). The data used as an input for GWAS can be found in **Table S2**.

## Genome-Wide Association Study

The genotype-specific mean was used as an input to GWAS. The GWAS was conducted as described in (Awlia et al. 2021 [\[5\]](#)), initially using 250 k SNPs (Horton et al. 2012 [\[6\]](#)) and subsequently using 4 M SNPs (1001genomes.org [\[7\]](#)). The GWAS associations were evaluated for minor allele count (MAC) and association strength above the Bonferroni threshold with  $-\log_{10}(\text{p-value}/\#\text{SNPs})$ , calculated for each sub-population of SNPs above threshold MAC (**Table S3**, Bonf.threshold.MAC.specific). The GWAS data was processed for associations with all SNPs used to generate Manhattan and QQ-plots (R notebook for processing GWAS data: <https://rpubs.com/mjulkowska/GWAS-root-shoot-Arabidopsis-salt-250kSNPs> [\[8\]](#)). Additionally, we prioritized the genetic loci where multiple SNPs were associated with individual traits, or multiple traits were associated with one genetic locus. Our final selection yielded 14 associations with traits measured under control conditions, while 9 and 23 associations were specific to traits measured under 75 and 125 mM NaCl, respectively (**Table S3**). The GWAS output files can be accessed at Zenodo Repository (GWAS using 250 k SNPs: <https://doi.org/10.5281/zenodo.7271751> [\[9\]](#); GWAS using 4 M SNPs: <https://doi.org/10.5281/zenodo.7271741> [\[10\]](#)). For the most promising loci (**Table S4**), the sequence information of 147 accessions belonging to the HapMap population was downloaded from the 1001 genomes project website (1001genomes.org [\[7\]](#)) and aligned with ClustalO as described in (Julkowska et al. 2016 [\[11\]](#)). More details on plotting the genome divergence in the loci of interest can be found at <https://www.protocols.io/view/plotdivergence-for-plotting-the-natural-variation-x54v98y84l3e/v1> [\[12\]](#). The input and output files of Plot Divergence analysis can be found in Zenodo repository (<https://doi.org/10.5281/zenodo.7278021> [\[13\]](#)).

## Haplotype analysis

For the most promising candidate loci (**Table S4**), we have identified the gene open reading frames that were located within the linkage-disequilibrium of the associated SNPs. The protein-coding genes were subsequently used for haplotype analysis. All the SNPs located within the gene coding region were used to perform a haplotype analysis. The detailed protocol used for haplotype analysis can be accessed at <https://www.protocols.io/view/haplotype-analysis-of-the-cds-in-arabidopsis-thali-5jyl86n8v2wp/v1> [\[14\]](#). The accessions were grouped into haplotype groups depending on the combination of SNPs within the protein-coding region of the gene of interest. The haplotypes represented by less than 3 accessions were excluded from further analysis. The phenotypes of individual haplotypes were explored for significant differences using ANOVA. The results for all investigated haplotypes can be accessed at Zenodo Repository (<https://doi.org/10.5281/zenodo.7277703> [\[15\]](#)).

## Identification of SR3G orthologs and paralogs

DUF247 domain-containing orthologs from *Arabidopsis thaliana* and eight other species were identified by reciprocal best BLAST within CoGe ((Nelson et al. 2018 [\[16\]](#)) <https://genomeevolution.org/CoGe/CoGeBlast.pl> [\[17\]](#)) using SR3G (At3g50160) nucleotide sequence as query with an E-value cutoff of  $1.00\text{E-}10$ . The gene ID, species, and gene names been used in this study are listed in **Table S5**. Sequence homologs were further confirmed by multiple alignments of those hits within each species using Geneious Prime. Sequences with missing or incomplete DUF247 domains were

removed, with the exception of the mis-annotated DUF427 homologs in *Camelina sativa* which contained an extraneous intron that, after removal, resulted in a set of intact DUF247 open reading frame containing genes.

## Phylogenetic and positive selection analysis of SR3G orthologs

Identified orthologs were translated into amino acid, then aligned by MUSCLE. Alignments were carefully validated by manual correction. Briefly, diverged insertions were trimmed, shifted frames were adjusted, and conserved regions after translation were retained for phylogeny reconstruction. Phylogenetic reconstruction was performed using the RAXML (Stamatakis 2014) plug-in within Geneious (Geneious Prime 2022.1.1) with the GAMMA-GTR nucleotide model, rapid bootstrapping with a parsimony search, and 100 bootstraps. The distribution of SR3G and other DUF247 orthologs, root lengths for each branch, were visualized by software Geneious Prime.

To characterize any potential positive selection that might exist along certain branches of the inferred phylogeny, including SR3G and its orthologs, we performed a branch-site model A test on branches leading to SR3G, and branches leading to the clade containing SR3G and its closet orthologs using PAML4.0 (Yang 2007) as performed previously (Beilstein et al. 2015; Nelson et al. 2014). Specific branches are denoted as  $\omega_0$ ,  $\omega_1$ ,  $\omega_2$ ,  $\omega_3$ , and  $\omega$ -rest. In the null hypothesis ( $H_0$ ): we assigned a substitution rate ( $\omega = dN/dS$ ) along with  $0 < \omega < 1$  or  $\omega = 1$  to all branches (represented by  $H_0$ :  $\omega_0 = \omega_1 = \omega_2 = \omega_3 = \omega$ -rest). The alternative hypotheses (HAs) here allow the differential selections been applied to respective branch(es). Specifically, the five alternative hypotheses are listed here:  $H_1$ :  $\omega$ -rest branches =  $\omega_0 = \omega_3 = \omega_1 \neq \omega_2$ ,  $H_2$ :  $\omega$ -rest branches =  $\omega_0 = \omega_1 = \omega_2 \neq \omega_3$ ,  $H_3$ :  $\omega$ -rest branches =  $\omega_0 \neq \omega_1 \neq \omega_2 \neq \omega_3$ ,  $H_4$ :  $\omega$ -rest branches  $\neq \omega_0 \neq \omega_1 \neq \omega_2 \neq \omega_3$ ,  $H_{sep}$ : each branch has its unique  $\omega$  value. To test these hypotheses, we further validate or reject alternative hypotheses by conducting a Chi-square test based on likelihood scores (Ln-L score) derived from the respective branch-site model A tests (Chi-square test cutoff:  $P < 0.05$ ).

## Screening candidate genes identified through GWAS

The genes located within the selected loci (Table S4) were further inspected for their contributions to root development and root:shoot ratio under control and salt stress conditions. The T-DNA insertion lines (Table S4) were genotyped using primers listed in Table S4, to confirm the homozygous insertion of the T-DNA within the phenotype lines. All the mutant lines were used for seed propagation in the controlled growth chamber with 16/8h of light/dark period, at 22°C, 60% humidity and 200  $\mu\text{mole m}^{-1} \text{s}^{-1}$  under long-day conditions. The seeds of plants identified as homozygous were subsequently used for the experiment on agar plates, as described before for the Genome-Wide Association Study. A detailed protocol for plate assays can be accessed at <https://www.protocols.io/view/plate-assay-for-identification-of-salt-stress-indu-eq2lyd1mwlx9/v1>. The plate images were analyzed for root system architecture using Smart Root (Lobet, Pagès, and Draye 2011) and the root:shoot ratio using the tool described before for Genome-Wide Association Study. The analysis of the mutant lines was performed in R. The data was first curated for selecting possible outliers within a genotype and subsequently used to fit growth functions for root and shoot tissues using exponential functions.

## Screening SR3G mutants within root:shoot locus

The following T-DNA lines (listed in Table S4) were used for SR3G (AT3G50160) as the main GWAS candidate: *sr3g-4* (SAIL\_690\_E12, glufosinate resistant) and *sr3g-5* (SAIL\_608\_C06, glufosinate resistant). All T-DNA insertion alleles were in the background of Col-0 accession. The position of T-DNA insertions was verified using PCR (primers listed in Table S4). RNAi lines targeting the neighboring of the SR3G or a combination of SR3G and its neighboring DUF247 genes included CSHL#3 targeting DUF130, CSHL#4 targeting SR3G, CSHL#5 targeting DUF130 and DUF170, CSHL#6 targeting DUF130 and DUF140, CSHL#7 targeting DUF10 and DUF150, CSHL#8 targeting DUF140 and DUF190, CSHL#10 targeting DUF180, CSHL#11 targeting DUF150 and DUF170, CSHL#12





we used SmartRoot plugin (Lobet, Pagès, and Draye 2011 [DOI](#)) in ImageJ to trace the root, and extracted root-related features in the CSV format followed by data analysis in R (<https://rpubs.com/mri23/913282> [DOI](#)).

## Evaluation of plant performance under salt stress in soil

The seeds were germinated in ½ MS media for one week, as described for the agar-based plate experiments. One week after germination, the seedlings were transplanted to the pot (12 x 4 cm insert) containing the Cornell Mix soil watered to 100% water holding capacity and placed in the walk-in growth chamber with the 16 h light / 8 h dark period, 22°C and 60% relative humidity throughout the growth period. When the pots dried to the weight corresponding to 50% of their water holding capacity, they were soaked for 1 h in tap water or a 200 mM NaCl solution, resulting in an effective concentration of 100 mM NaCl based on the 50% soil water holding capacity, which corresponded to a moderate level of salt stress (Awlia et al. 2016 [DOI](#)). We then allowed the pots to be drained for 2-3 h to eliminate excess moisture. The pots were placed under phenotyping rigs equipped with an automated imaging system (L. 'ang Yu et al. 2023 [DOI](#)) and the pot weight was measured daily to maintain the reference weight corresponding to 50% of the soil water holding capacity throughout the experiment.

To evaluate the growth of soil-grown plants, we used an in-house developed automated imaging system using Raspberry Pi cameras to capture the growth dynamics over the period of two weeks (L. 'ang Yu et al. 2023 [DOI](#)). The collected images were processed using PlantCV pipeline (Gehan et al. 2017 [DOI](#)), followed by the data analysis in R, as described here ([https://rpubs.com/mjulkowska/sr3g\\_Maryam2021OctNov](https://rpubs.com/mjulkowska/sr3g_Maryam2021OctNov) [DOI](#)).

## Transient expression of SR3G constructs in tobacco epidermal leaf

*Agrobacterium tumefaciens* strain GV3101 containing cellular marker free GFP (35S::GFP with rifampicin and spectinomycin resistant), generously provided by Professor Maria Harrison Lab at Boyce Thompson Institute, or mVenus-SR3G constructs (including UBQ10::mVenus-SR3G-full length, UBQ10::mVenus-SR3G-fragment A, UBQ10::mVenus-SR3G-fragment B, and UBQ10::mVenus-SR3G-fragment AB, all with spectinomycin and tetracycline resistant) were grown in liquid YEP medium with appropriate antibiotics overnight at 28°C while shaking. The following concentrations were used for antibiotics when used for transient transformation: spectinomycin (50 ug/ml), tetracycline (10 ug/ml), and rifampicin (12.5 ug/ml). Rifampicin and kanamycin (50 ug/ml) were used for P19 plasmid (generously provided by Maria Harrison Lab at BTI) as a silencing inhibitor. The following day, OD<sub>600</sub> was measured to be 0.3-0.5. The cultures were then centrifuged at 8000x g for 2 min. The *Agrobacterium* pellet was resuspended in an induction buffer containing 10 mM MgCl<sub>2</sub>, 10 mM MES, pH 5.6, and supplemented with 200 µM acetosyringone and was incubated for 2-4 h in the dark at room temperature. Leaves of 4-weeks-old *Nicotiana benthamiana* plants were infiltrated using a syringe. After 48 hours, the fluorescence was visualized in epidermal cells of leaf discs using confocal microscopy.

Confocal microscopy images were taken using the Leica TCS SP5 Laser Scanning Confocal Microscope with the following setting: Excitation at wavelengths of 488 nm (GFP), 514 nm (mVenus), and 561 nm (chlorophyll autofluorescence) were provided using an argon or DPSS (Diode-Pumped Solid State) laser. A spectral emission range of 505–525 nm for GFP, 526–560 nm for mVenus, and 680–700 nm for chlorophyll autofluorescence was used.

## Elemental analysis

The elemental analysis was performed using Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP-AES). For analysis of Na<sup>+</sup> and K<sup>+</sup> ions in roots and shoots of plate-grown seedlings, the plants were grown on ½ x MS plates as described above. After two weeks of exposure to 0, 75 or 125 mM NaCl, root and shoot tissues were harvested and measured for fresh

weight, rinsed in milli-Q water and collected into separate paper bags that were dried at 60 °C for 2-3 days. Subsequently, the dry weight was recorded. Samples were digested in double distilled HNO<sub>3</sub>, followed by the addition of 60/40 nitric/perchloric acid continuing incubation at 150 °C. Samples were processed for ICP-AES analysis using a Thermo iCap 7000 ICP-AES after diluted to 10 ml with deionized water. For soil-grown plants, the entire rosette leaves were harvested at the end of week four and underwent a similar procedure as described above. Ion content was calculated per dry weight for each sample, followed by data analysis in R (sr3g:<https://rpubs.com/mri23/913285> , wrky75/sr3g double mutants: <https://rpubs.com/mri23/1157092>  for plate-grown and sr3g:<https://rpubs.com/mri23/913299>  for soil-grown seedlings).

## Electrolyte leakage analysis

The electrolyte leakage measurement was done as previously described (Ying Hu et al. 2015 ) with slight modifications. In brief, for each plant, three leaf discs were incubated in 2 ml of distilled water for 24 h at the light at room temperature with gentle shaking, followed by measuring the initial electrolyte leakage using a conductivity meter (Horiba LAQUAtwin EC-11 Compact Conductivity Meter, # 3999960125). The samples were then subjected to 80 °C for 2 h to release the total electrolyte, and cooled at room temperature for 24 h. Final electrolyte leakage was measured the following day. The electrolyte leakage percentage was calculated by dividing initial conductivity by final conductivity \* 100 and followed by data analysis in R (<https://rpubs.com/mri23/913288>  and <https://rpubs.com/mri23/1102053> ).

## RT-qPCR analysis

Total RNA was isolated from Arabidopsis seedlings root and shoot treated with or without 75 or 125 mM NaCl for one week using the RNeasy Plant Mini Kit (Qiagen). RNA samples were cleaned at least twice using RNase-free DNase (Qiagen) to remove genomic DNA contamination. One microgram of total RNA was used for a reverse transcriptase reaction using iScript cDNA Synthesis Kit (Catalog #170-8891; Bio-Rad). RT-qPCR analysis was conducted using iQ SYBRGreen Supermix (Bio-Rad) according to manufacturer's instructions in the CFX96 real-time PCR system (Bio-Rad). *AT4G04120* (*transposable\_element\_gene*) was used as a housekeeping gene for data normalization. RT-qPCR experiments were conducted using three independent experiments, each with at least three biological replicates. The list of primers used is shown in **Table S4**.

## Suberin staining

Suberin content was examined in 6-days-old Arabidopsis roots treated with or without 75 mM NaCl for two days using Fluorol Yellow 088 (FY; Santa Cruz Biotechnology Incorporation) staining. Seedlings grown on agar plates (as described above for mutant screens) were incubated in a freshly prepared solution of FY (0.01% [w/v] in lactic acid 85% [W/W]) at 70°C for 30 min, rinsed in milli-Q water for three times with 5 min each, and mounted on slides with 50% glycerol before confocal microscopy examination using excitation at wavelength of 488 nm and a spectral emission range of 500–550 nm. Suberin quantification was performed using the ImageJ software followed by data analysis in R (<https://rpubs.com/mri23/935019> ).

## Suberin monomer quantification using Gas Chromatography–Mass Spectrometry (GC-MS)

100 - 150 Arabidopsis seedlings were grown on the plates for four complete days, transferred to 75 mM salt plate for two days, and roots were harvested in distilled water to remove any remaining agar. Each plate was treated as one biological replicate, and five plates were used for each genotype/condition. Each plate of roots was collected into a microcentrifuge tube and dried in a speed vacuum concentrator (Thermo Electron, Waltham, MA), and weighed. Approximately 1 mg of dried root were collected from each plate. Suberin monomer quantification was performed according to (Delude et al. 2017 ; Kreszies et al. 2019 ) with modifications. Dried root tissue was



incubated in 1% cellulase plus 1% pectinase in 10 mM citrate buffer (pH 3.0) for two weeks, with the incubation solution changed every three days. After the cellulase/pectinase solution was removed, chloroform/methanol solution (2:1 and 1:2, v/v) were mixed with the root tissue for three days respectively, to remove lipids. The remaining root tissue was mixed with 1 ml freshly prepared 5% H<sub>2</sub>SO<sub>4</sub> in methanol (v/v) and 200 µl of toluene containing 20 µg of heptadecanoic acid as internal standard, sealed in an 8 ml capped tube and heated at 85°C for 90 min. After samples were cooled to room temperature, 200 µl hexane and 1.5 ml 0.9% sodium chloride in water are added to the capped tube with shaking. After centrifuging at 1000 g for 1 min, the upper organic phase was transferred to a glass insert of GC vial and dried. 40 µl pyridine was added to the glass insert, 40 µl *N,O*-bis(trimethylsilyl)trifluoroacetamide (Restek, Bellefonte, PA) was added subsequently, incubated at 65°C for 2 h. Standards of 1 µl derivatized sample was injected into GCMS (Agilent 6890-5975, Santa Clara, CA), with the inlet temperature of 250°C, column helium flow of 1 ml/min, and a split ratio of 5:1. The oven temperature was held at 50°C for 1 min, raised to 200°C at the rate of 25°C/min, held at 200°C for 1 min, raised to 320°C at the rate of 10°C/min, and finally held at 320°C for 5 min. 1-Docosanol (22:0-OH), 1-eicosanol (20:0-OH), hexadecanedioic acid (16:0-DCA), 1-hexadecanol (16:0-OH), 16-hydroxy hexadecanoic acid (16:0-ωOH), C4 - C24 Even Carbon Saturated FAMES (Sigma 49453-U), 20-hydroxy arachidic acid (20:0-ωOH) (Cayman Chemical Company, Ann Arbor, Michigan) standards were input at 2.5 µg, 5 µg, 10 µg, and 20 µg into a mixture and processed with the same procedure as the tested samples to build the standard curve. The MS spectrum of derivatized products with ±2n carbon number of these standards was calculated and identified, referring to the retention order in (Delude et al., 2017 [DOI](#)).

## Yeast One-hybrid Assay

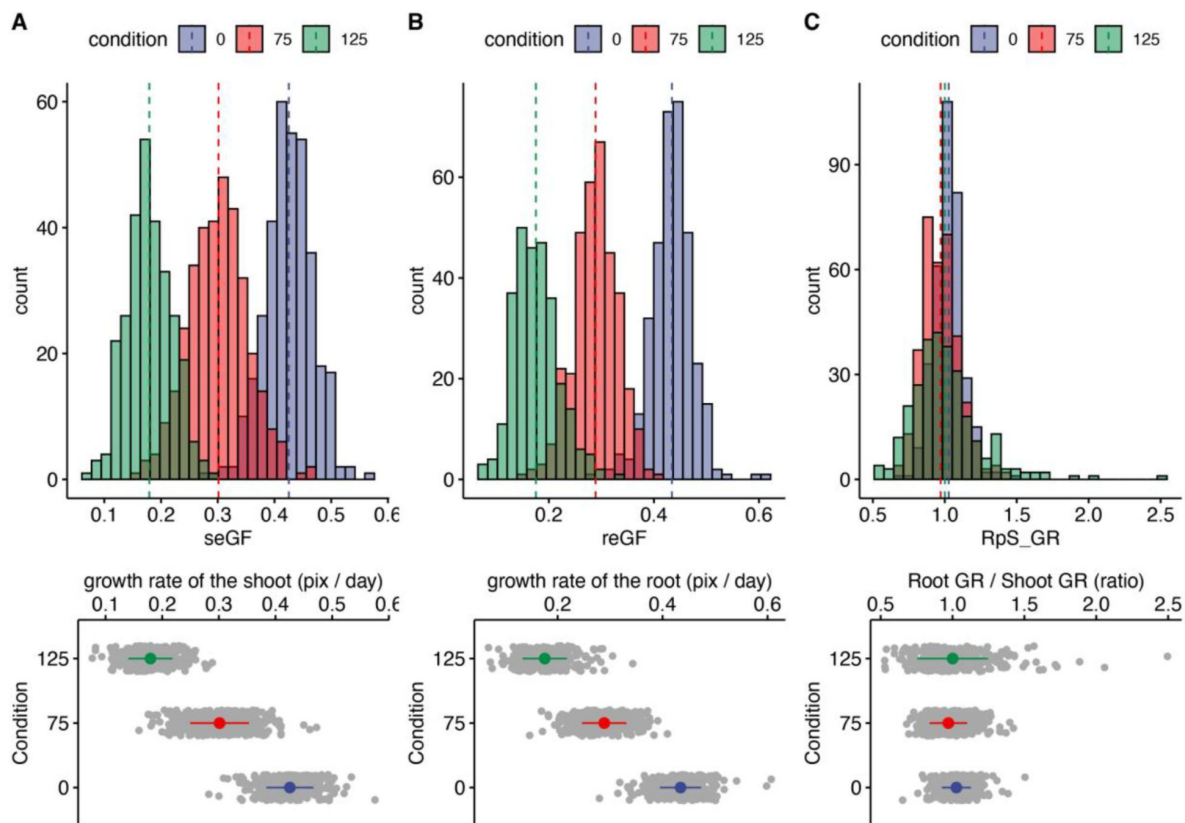
The respective combinations of *pB42AD*-fusion effectors and *pLacZ* reporters were co-transformed to yeast strain EGY48 (Shanghai Maokang, China). Transformants were selected and grown on SD/-Trp-Ura medium. The putative interactions were observed on SD/-Trp-Ura + X-Gal after a 3d-incubation at 30°C. Yeast transformation and growth assay were performed as described in the Yeast Protocols Handbook (BD Clontech, USA). In brief, 631 bp of the *DUF247-150* (AT3G50150) and 953 bp of the *SR3G* (At3g50160) promoter region were separately ligated into *pLacZ* vector as the reporter, while the CDS of *WRKY75* (AT5G13080) was cloned into *pB42AD* vector as the effector. The *SR3G::pLacZ* and *WRKY75-pB42AD* were transformed into the yeast strain EGY48 (Shanghai Maokang, China). The respective combinations of *pB42AD*-fusion effectors and *pLacZ* reporters were co-transformed into yeast strain EGY48. Transformants were selected and grown on SD/-Trp-Ura medium. The putative interactions were observed on SD/-Trp-Ura + X-Gal after a 3d-incubation at 30°C. Yeast transformation and liquid assay were conducted as described in the Yeast Protocols Handbook (BD Clontech, USA). The primers used for the Y1H assay are shown in **Table S4**.

## Results

### Natural variation in salt stress-induced changes in root:shoot ratio

To investigate salt stress-induced changes in root:shoot ratio, we developed a custom tool that quantifies green and white pixels, corresponding to shoot and root of Arabidopsis seedlings grown on agar plates (**Fig. 1** [DOI](#)). The projected area of root and shoot showed high correlation with root and shoot fresh weight (**Fig. S1**), indicating that projected root and shoot area can be used as an accurate proxy to non-destructively measure the increase in root and shoot biomass from the images of seedlings grown on agar plates.

Using our custom tool, we re-analyzed the data consisting of 360 Arabidopsis accessions grown on plates supplemented with 0, 75, and 125 mM NaCl (Julkowska et al. 2017 [DOI](#)), for salt stress-induced changes in the root:shoot ratio (**Fig. 1** [DOI](#), **Fig. S2**). The growth of root and shoot for each plant was estimated using the exponential function. Salt stress-induced reduction in growth rate was



**Figure 1.**

**Salt stress is causing discoordination of root and shoot growth. Arabidopsis HapMap population was screened for salt stress-induced changes in root:shoot ratio.**

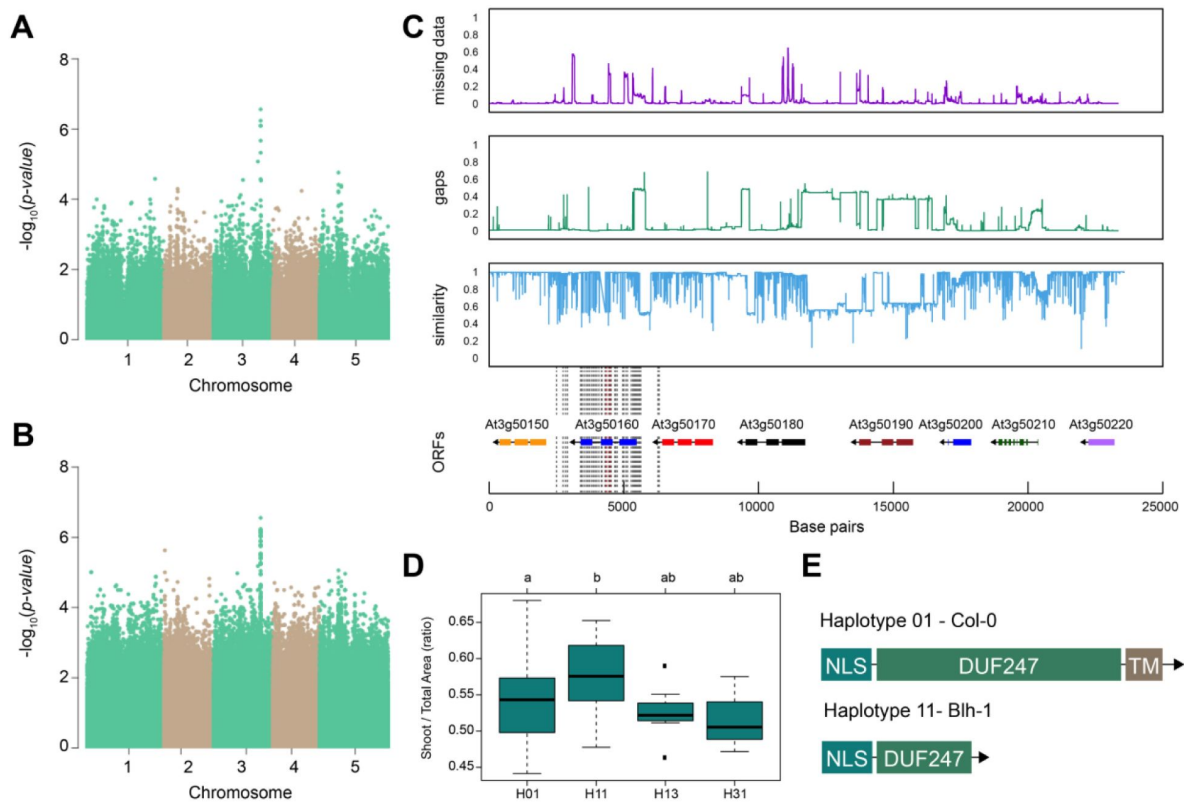
The increase in the projected area of shoot and root (Fig. S2) was used to estimate growth rate of (A) shoot and (B) root using an exponential function. (C) The root:shoot growth rate ratio was calculated per genotype. The histograms represent the number of accessions across three studied conditions (0, 75 and 125 mM NaCl), whereas the population average is indicated using the dashed line. Additionally, the distribution of the genotypes within each treatment was additionally visualized using the error plots (lower panel), where population average and standard error is indicated using a colored point and a line respectively. Individual grey points represent individual genotypes.

observed to be dose-dependent for both root and shoot (**Fig. 1A-B**). In general, the root growth was more sensitive to salt stress compared to shoot growth, with the population-wide relative growth rates reduced to a fraction of 0.67 and 0.41 of the growth rates observed at control for 75 and 125 mM NaCl, respectively. In comparison, the growth rates of the shoot were reduced to 0.71 and 0.43 of the control in 75 and 125 mM NaCl treatments, respectively (**Fig. S3**). While the mean value of root:shoot growth rate did not change upon salt stress treatment, the variance in the root:shoot ratio expanded with the increasing concentrations of salt (**Fig. 1C**). These results suggest that while root and shoot growth are well coordinated under non-stress conditions, salt stress exposure results in loss of coordination of organ growth across *Arabidopsis* accessions. No significant correlation was found between the root:shoot growth ratio and total seedling size (**Fig. S4, S5**), indicating that the relationship between coordination of root and shoot growth and salt tolerance during the early seedling establishment is complex.

## Identification of genetic components underlying salt-induced changes in root:shoot ratio

To identify genetic components underlying salt-induced changes in root:shoot ratio, we used the collected data as an input for GWAS. The associations were evaluated based on the p-value, the number of SNPs within the locus, and the number of traits associated with individual loci. As Bonferroni threshold differs depending on the minor allele count (MAC) considered, we identified significant associations based on a Bonferroni threshold for each subpopulation of SNPs based on MAC (**Table S3**). The initial GWAS was performed using 250 k SNPs (Horton et al. 2012), and led to the identification of 14 significantly associated loci, of which 3 and 7 loci were associated with traits measured at 75 or 125 mM NaCl treatments, respectively. We identified one salt specific association on chromosome 2 (**Fig. S6**), where two SNPs were identified within AT2G27300 encoding NAC-domain containing transcription factor (NTL8). Additionally, salt-induced changes in root:shoot ratio and shoot:total seedling area were associated with SNPs on chromosome 3, where 2 significantly associated SNPs were flanking AT3G22430, encoding an unknown gene encoding a domain of unknown function (**Fig. S7**). To further prioritize the selection of candidate genes for further validation study, we repeated GWAS using 4 M SNPs (1001 Genomes Consortium. Electronic address: magnus.nordborg@gmi.oeaw.ac.at and 1001 Genomes Consortium 2016). Increased SNP density GWAS revealed 32 additional loci, with only one significantly associated locus being picked up by both 250k and 4M SNPs GWAS (locus 30, **Table S3**). We further prioritized the identified associations based on the number of significantly associated SNPs, minor allele count (MAC), and number of traits associated with each locus, and identified three loci of major interest (**Table S4**). The region within linkage disequilibrium (LD) of the significantly associated SNPs was investigated for the sequence divergence and open reading frames (**Fig. 2**, **Fig. S8**, **Fig. S9**).

Out of identified associations, locus 30 stood out as it consisted of nine SNPs associated with p-value above the Bonferroni threshold, and was associated with both root:shoot and shoot:total seedling area at 6 and 8 days after transfer of the seedling to 125 mM NaCl (**Fig. 2A-B**, **Fig. S8**, **Table S4**). The significantly associated SNPs were located in and around the gene coding region of AT3G50160, which encodes a domain of unknown function 247 (DUF247). We investigated the region for sequence divergence, and found that the region contained not only a high degree of sequence diversity but also potential structural variants (**Fig. 2C**). To identify allelic variation underlying the initial associations, we performed haplotype analysis, where accessions with shared SNP pattern within the protein-coding part of AT3G50160 were grouped into individual haplotype-groups (**Fig. 2D**). Each unique haplotype, represented by at least three accessions, was subsequently studied for differences in observed phenotypes. We observed that Haplotype 1, represented by 46 accessions including Col-0, was significantly different from Haplotype 11, represented by 11 accessions including Blh-1, in shoot-per-total seedling area at 6 days after transfer to 125 mM NaCl (**Fig. 2D**), as well as shoot growth rate and relative shoot growth rate at 125 mM NaCl (**Fig. S9**). Upon further investigation and sequencing of AT3G50160 in both Col-0 and



**Figure 2.**

### SR3G is associated with salt stress-induced changes in shoot-per-total seedling area ratio.

The salt induced changes in root-to-shoot and shoot-per-total seedling area ratios were used as an input for GWAS. Manhattan plots representing the associations of shoot-per-total seedling area recorded 6 days after transfer to 125 mM NaCl and (A) 250 k SNPs and (B) 4 M SNPs. (C) Significant associations were found with the SNPs forming a locus on chromosome 3 in and around *SR3G* (AT3G50160), encoding Domain of Unknown Function 247 (DUF247). The natural variation in the LD region was studied in 162 accessions sequenced by 1001 Genomes Project. The upper panel represents portion of the missing data, upper middle panel represents deletions relative to Col-0, while lower middle panel represents the sequence similarity compared to Col-0. The bottom panel represents the open reading frames (ORFs) within the LD, and the location of associated SNPs is indicated with the dashed lines. Red dashed lines represent associations above Bonferroni threshold in 250k SNP mapping, while grey lines represent associations with  $-\log_{10}(p\text{-value}) > 5$  in the 4M SNP mapping. (D) The haplotype analysis performed using SNPs located within the coding region of the *SR3G* revealed significant differences in shoot-per-total seedling area recorded 6 days after transfer to 125 mM NaCl between Haplotype groups 1 (represented by 46 accessions, including Col-0) and 11 (represented by 11 accessions, including Blh-1). The significant differences between individual haplotype groups were tested using ANOVA with Tukey HSD to identify significantly different groups. (E) Upon further sequencing of *SR3G* in accessions from Haplotype 1 and 11, revealed two 200 bp insertions within *SR3G* exons in 4 tested accessions belonging to haplotype 11 group. These 200 bp insertions resulted in a premature STOP-codon within the DUF247 domain (after Gly-215).

Blh-1 accessions, we discovered that the Blh-1 allele carries a large insertion at 644 nucleotides downstream from the ATG of AT3G50160, followed by multiple other sequence variations (**Fig. S10**). The first insertion leads to a premature STOP-codon within the DUF247 domain after Gly-215 (**Fig. 2E**, **Fig. S11**), resulting in a truncated protein that still carries a nuclear localization signal and coiled-coil domain, but is missing a major part of the calcium-binding domain and a predicted transmembrane domain (**Fig. S12**). As there are other DUF247s within the locus associated with salt-induced changes in root:shoot ratio, we decided to name the gene containing majority of the associated SNPs (At3g50160) as Salt Root:shoot Ratio Regulator Gene (SR3G). Together, these results suggest that the allelic variation in SR3G protein sequence might underlie the observed differences in root:shoot ratio under salt stress conditions.

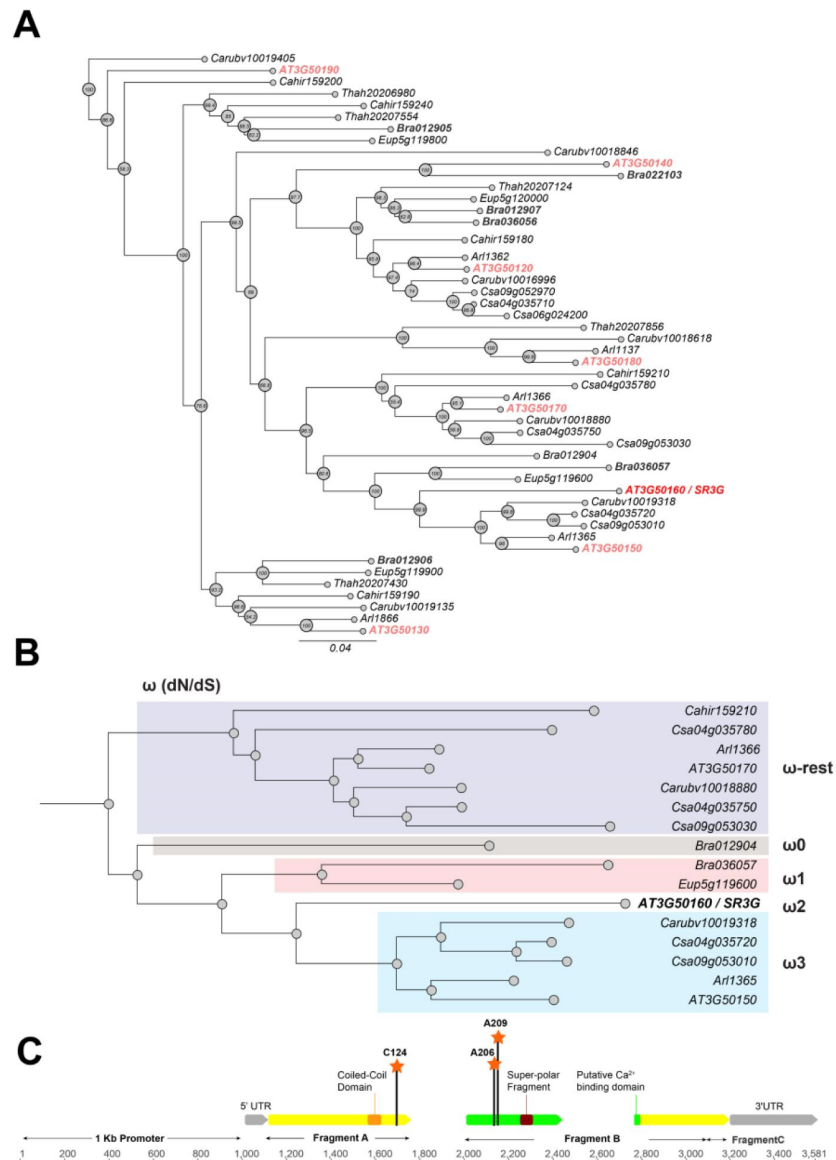
## Evolutionary context of SR3G locus reveals its origin in local genome duplication

SR3G sits in what appears to be a tandemly duplicated array of DUF247s along Arabidopsis chromosome 3. The genome-wide identification of SR3G included eight DUF247 paralogs, AT3G50120, AT3G50130, AT3G50140, AT3G50150, AT3G50170, AT3G50180, AT3G50190, and AT3G50200. Reciprocal best BLAST within CoGe's BLAST tool of these nine DUF247 paralogs against the genomes of seven other Brassicaceae (see **Table S5**) revealed a total of 50 potential orthologs used for downstream analysis. While the copy number of DUF247 loci are similar across the examined species, the noted exceptions are those with recent whole genome duplication events (*Brassica rapa* and *Camelina sativa*). Given these data, it appears that DUF247 loci are commonly tandemly duplicated, although the eight paralogs observed in Arabidopsis represents an extreme case for this gene family.

To examine conservation within the predicted DUF247 functional domains, we generated an amino-acid based multiple sequence alignment using MUSCLE. All of the identified DUF247 proteins contained the DUF247 domain near the center of the translated sequence. While partial deletions were identified outside of the DUF247 domain at the C-terminus of a few DUF247s, such as *CARHR159180*, *Csa09g052970*, and *Bra01902*, in general the DUF247 domains were highly conserved across Brassicaceae. The amino acid sequences spanning the “coiled-coil domain” (Amino acid position: Met153-Lys173), putative  $\text{Ca}^{2+}$  binding domain (Amino acid position: Gly210-Phe372), and super polar fragment (Amino acid position: Ser301-Ile320) were further selected for phylogenetic reconstruction (**Fig. 3B**). The resulting phylogeny demonstrated that SR3G is sister to a clade containing AT3G50150 and representatives from *C. sativa*, *Capsella rubella*, and *A. lyrata*, with representative DUF247s from *B. rapa* and *Eutrema salsugineum* serving as a monophyletic outgroup. These data support a hypothesis where SR3G is a recent duplicate (likely A/C Clade specific; (Forsythe, Nelson, and Beilstein 2020)) of AT3G50150 and that representative SR3G sequences in *C. sativa*, *C. rubella*, and *A. lyrata* have been lost.

Given the specific retention of AT3G50150 in *A. thaliana*, we next tested whether this locus is experiencing selection. To test this assumption, we grouped the branches leading to SR3G into four groups (**Figure 3B**:  $\omega_0$ ,  $\omega_1$ ,  $\omega_2$ , and  $\omega_{\text{rest}}$ ) and then performed a branch-site model A hypothesis test. In our null hypothesis ( $H_0$ ), we hypothesized that all branches have the same selection pressure ( $H_0$ :  $\omega_{\text{rest}} = \omega_0 = \omega_1 = \omega_2$ ). For our alternative hypothesis, we created five different  $\omega$  groups to cover scenarios where specific branches may be experiencing selection (See Methods). Based on our analysis, the branch leading to SR3G is experiencing selection (likelihood score = 20625.5, Chi-square derived P value;  $P = 0.00418$ ). Interestingly, no other branches appeared to be experiencing selection. To pinpoint specific amino acids within SR3G under selection, we performed the Bayes Empirical Bayes (BEB) test, identifying three sites with a posterior probability > 0.90, including 124C ( $P = 0.992$ ), 206A ( $P = 0.951$ ), and 209A ( $P = 0.988$ ). Two out of three sites are located at the putative  $\text{Ca}^{2+}$  binding site and all three sites possess a unique amino acid compared with the other 49 amino acid sequences (**Fig. 3C**).





**Figure 3.**

### Phylogeny and positive selection analysis of SR3G and its orthologs from other species.

The orthologous genes of SR3G (AT3G50160) from other seven species were identified using GoGeBlast (E-value < 1.00E-10). **(A)** The protein sequences of the 50 homologous genes were aligned by MUSCLE and then an unrooted phylogenetic tree was reconstructed by a Maximum-likelihood (ML)-derived style by RAXML (Bootstrap number: 100). **(B)** A positive selection analysis was conducted by branch-site model A test: Specifically, we hypothesized the differential or constant dN/dS ( $\omega$ ) substitution rate among the closely distant branches leading to DUF247 (AT3G50160) (See Method & Materials for details). These branches were marked as  $\omega_1$  (light brown branch),  $\omega_2$  (DUF247-specific branch),  $\omega_3$  (light blue branch), and  $\omega$ -rest (light purple branch). **(C)** The gene structure of DUF247 (AT3G50160) was illustrated by wide arrows (grey arrow: 5 and 3' UTR, yellow arrow: coding-region). The regions used for cloning (Fragment A, B, and C) were marked by narrow arrows along with "Coiled-Coil Domain", "Putative Ca<sup>2+</sup> binding domain" and "Super-polar Fragment" marked on Fragment A and B, respectively. Lastly, the Bayes Empirical Bayes (BEB) was performed to test the probability of sites along with  $\omega > 1$  over the DUF247 (AT3G50160). Three sites were generated with a posterior probability > 0.90: 124C (P = 0.992), 206A (P = 0.951), and 209A (P = 0.988). These three sites were marked with an orange asterisk and assumed to have been undergoing positive selection.

## SR3G expression is enhanced in root by salt stress

To identify the developmental window and specific plant tissues where *SR3G* gene is active, we examined its expression pattern. Based on Arabidopsis eFP Browser (<http://bar.utoronto.ca/>), *SR3G* expression is relatively low, and restricted to the early stages of seedling development (Fig. S13A). Upon examining the expression profiles specific to different tissues, we observed that *SR3G* is predominantly expressed in the root stele (Fig. S13B). Moreover, *SR3G* expression was induced by salt stress (Fig. S13C). To validate these results, we performed RT-qPCR on 11 days old seedlings of Col-0 exposed to 0 or 75 mM NaCl for 7 days. In Col-0 seedlings, the transcript levels of *SR3G* were higher in shoot compared to the root tissue (Fig. 4A). Interestingly, the *SR3G* transcripts increased only in root (54 folds) upon salt stress exposure but not in shoot (Fig. 4A). These results suggest that *SR3G* has a root-specific function in the salt stress.

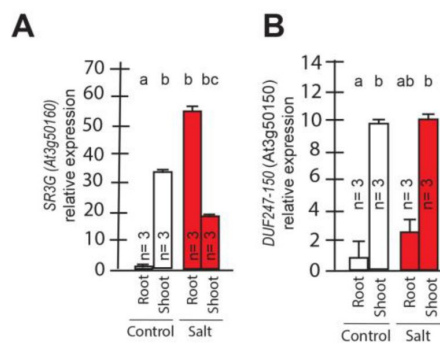
We further examined expression profiles of other *DUF247* genes that directly neighbor *SR3G* in response to salt stress using the eFP Browser. We observed similar expression patterns as *SR3G* for other *DUF247*s including At3g50150, indicating their responsiveness to salt stress (Fig. S14). Because *DUF247-150* (At3g50150) is closely located to the *SR3G* in chromosome 3, we investigated its gene expression in young seedlings. We observed that its expression was higher in shoot compared to the root tissue. However, its expression remained unchanged in both root and shoot under salt stress (Fig. 4B). For the remaining *DUF247* genes (At3g50120 and At3g50200), we observed no significant changes, while others (At3g50130, At3g50140, At3g50170, At3g50180) showed reduced expression upon exposure to salt stress (Fig. S14). These results suggest that *SR3G* is the main gene involve in regulation of root development under salt stress.

## SR3G relocates from plasma membrane to nucleus/cytosol after removal of its transmembrane domain

*SR3G* protein is predicted to contain a nuclear localization signal (Lys15-Ala19), a coiled-coil domain (Met153-Lys173), a  $\text{Ca}^{2+}$  binding pocket (Gly210-Phe372), and a trans-membrane domain (Pro469-Thr503) (Fig. S15). We generated constitutive expression constructs of full-length *SR3G* gene CDS, and a combination of individual domains (refer to as A, B, and AB domains) to identify their subcellular localization (Fig. S15). Fragment A comprised a nuclear localization signal, a coiled-coil domain, and the initial segment of the  $\text{Ca}^{2+}$  binding pocket, encompassing seven amino acids. Fragment B encompassed the remaining segment of the  $\text{Ca}^{2+}$  binding pocket. Fragment C only included the transmembrane domain. All constructs contained N-terminal fusion of mVenus. The individual constructs were used in transient expression assay in tobacco leaves. Our result showed that full-length *SR3G* localizes to the plasma membrane (Fig. 5A) as its signal overlapped with FM4-64, the plasma membrane marker (Fig. 5B). The *SR3G* fragments from which the transmembrane domain was excluded—namely fragments A, B, and AB— exhibited localization in both the nucleus and cytosol, resembling the pattern observed with free-GFP localization (Fig. 5C-F). It is noteworthy that while the signal from fragment A was directed to both the cytosol and nucleus, its intensity was notably stronger in the nucleus (Fig. 5C), aligning with the prediction that it contains the nuclear localization signal. Together, these findings indicate that the positioning of *SR3G* within the plasma membrane is due to the transmembrane domain encoded within the fragment C, and its removal could potentially release the protein into both the nucleus and cytosol.

## Lower expression of *SR3G* results in reduced lateral root development under salt stress

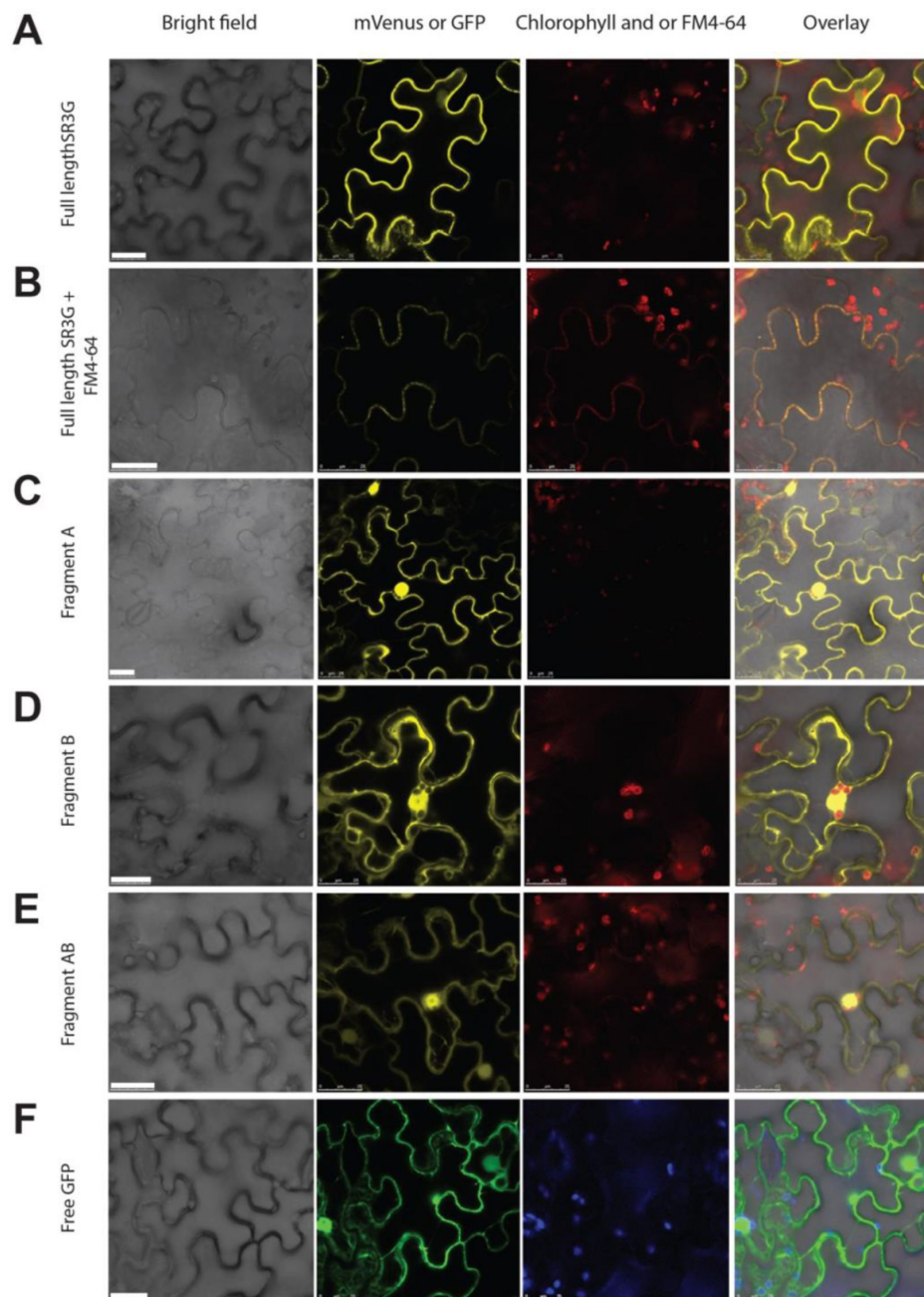
To investigate role of *SR3G* in salt-induced changes in root growth and development, we studied the root system architecture of two T-DNA insertion lines: *sr3g-4* (SAIL\_690\_E12) and *sr3g-5* (SAIL\_608\_C06) localized in second exon and first intron region of the *SR3G* gene, respectively (Fig.



**Figure 4.**

#### Transcript abundances for the two closely related *DUF247s*.

Expression of **(A)** *SR3G* (AT3G50160) **(B)** and its closest homologue *DUF247-150* (AT3G50150) were measured in Col-0 seedlings grown with and without salt stress. RT-qPCR analyses were conducted using seedlings grown on 1/2x MS for 4 d and then followed by transferring to the treatment plates with or without 75 mM NaCl for one week. Mean values are shown  $\pm$  SE, with number of replicates (n) shown in each graph. *AT4G04120* was used as a reference gene for normalization. Significance was determined by the Tukey-Kramer HSD test in JMP.



**Figure 5.**

**SR3G full-length protein resides in the plasma membrane while removal of its transmembrane domain results in protein relocation to nucleus and cytosol.**

SR3G full-length protein (A-B) or truncated versions (C-E) fused to mVenus at the N-terminus were agro-infiltrated into the *Nicotiana benthamiana* leaves for transient expression. Shown are a bright-field image of transfected leaf cells and mVenus-SR3G-mediated fluorescence as well as GFP-based subcellular marker. (A) Localization of full-length SR3G alone or (B) in combination with the known plasma membrane dye, FM4-64. Localization of truncated versions of SR3G are shown for (C) SR3G-Fragment A, (D) SR3G-Fragment B, and (E) SR3G-Fragment AB. (F) Free GFP was used as a nuclear and cytosolic marker. Scale bar = 25  $\mu$ m.

**S16A**). Among these two mutant lines, *SR3G* expression was significantly reduced in the *sr3g-5* mutant compared to Col-0 (**Fig. S16B**), indicating the *sr3g-5* is a knockdown mutant. We did not observe any differences in *SR3G* transcript abundance between *sr3g-4* and Col-0 (**Fig. S16B**). Moreover, we examined the transcript abundance of the closely related gene (i.e., *DUF247-150*, *At3g50150*) in the *sr3g-5* mutant, and found no significant difference. In the case that *DUF247-150* would act redundantly to *SR3G*, we would expect that the expression of *DUF247-150* would be increased in *SR3G* knockdown mutant. As this is not the case, these results further contribute to the evidence that *SR3G* and *DUF247-150* are not acting redundantly (**Fig. S16C**).

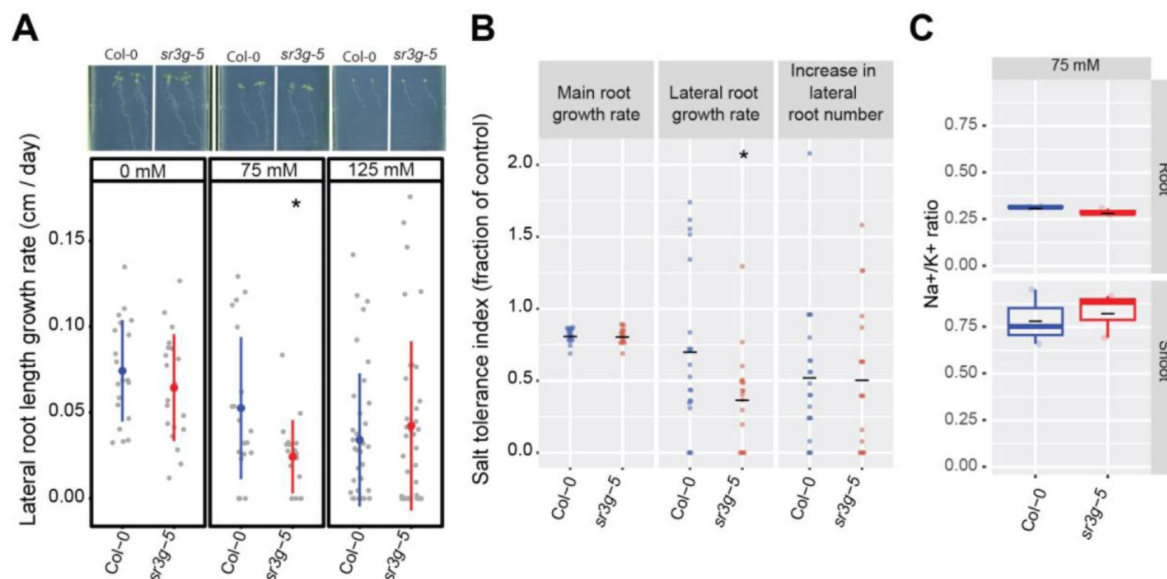
Two T-DNA insertion lines were subsequently studied for salt-induced alterations in root architecture (**Fig. 6** and **Fig. S17 and S18**). Among the various features of the root system architecture that we investigated (**Fig. S17**), *sr3g-5* exhibited a significantly reduced growth rate of lateral roots under 75 mM NaCl, but not under 125 mM NaCl, when compared to Col-0 (**Fig. 6A**). Likewise, the *sr3g-5* mutant displayed a significant reduction in the Stress Tolerance Index (STI), calculated as the ratio of growth rate under salt to growth rate under control conditions, for lateral root length under 75 mM NaCl (**Fig. 6B**). No significant difference was observed between *sr3g-4* and Col-0 (**Fig. S18**), except for Stress Tolerance Index (STI) calculated using growth rates of lateral root length and number, consistent with the observations made for the *sr3g-5* mutant line (**Fig. 6B**). Based on the RT-qPCR results (**Fig. S16**), we proceeded to further investigate the role of *SR3G* in salt stress, focusing on the *sr3g-5* mutant in our subsequent investigations.

As  $\text{Na}^+$  exclusion from the shoot is one of the main mechanisms of salt tolerance (Munns and Tester 2008; Møller et al. 2009; Julkowska et al. 2017), we further investigated the role of *SR3G* in ion exclusion using plate-grown seedlings. We used root and shoot samples collected for Col-0 and *sr3g-5* mutant to examine  $\text{Na}^+$  and  $\text{K}^+$  accumulation using ICP-AES at 14 days after exposure to salt stress. While salt stress increased  $\text{Na}^+$  and decreased  $\text{K}^+$  in all studied samples (**Fig. S17**), we observed no significant differences between Col-0 and *sr3g-5* mutant for  $\text{Na}^+$  and  $\text{K}^+$  accumulations and their corresponding ratio (**Fig. 6C**). These results suggest that *SR3G* plays a role in root development under the salt, but does not affect ion accumulation at early developmental stages.

To evaluate the effects of other closely related DUF247 genes on root development, we have investigated the T-DNA insertion lines targeting the neighboring DUF247 genes (AT3G50120, AT3G50130, AT3G50140, AT3G50150, AT3G50170, AT3G50180, AT3G50190, AT3G50200), and RNAi lines targeting one or more DUF247s within the interval (**Table S4, Fig. S19-S23**). For the screened T-DNA insertion lines (**Fig. S19-20**), we have not observed consistent changes in main or lateral root development in either control or salt stress conditions across alleles targeting the same gene. The only exception was *DUF247-200* encoded by AT3G50200, where two out of three studied T-DNA insertion alleles exhibited increased salt tolerance index for main root growth rate (**Fig. S20**).

For the screened RNAi lines (**Table S4, Fig. S21-S23**), we observed the most interesting root architecture phenotypes in lines targeting the neighboring genes of the *SR3G* (RNAi line #04, targeting AT3G50150, RNAi line #16 targeting AT3G50130, *SR3G*, and AT3G50170, and RNAi line #19 targeting AT3G50130, AT3G50140, AT3G50150, *SR3G*, AT3G50170, AT3G50180, and AT3G50190). Three out of five screened independent transformation lines from RNAi line #04 showed a decreased relative main root length (Salt / Control, CSHL #4 B2a, D4a, and G1d), whereas two lines showed a decrease in relative lateral root number (CSHL #4 D4a and F1b), and one line showed decreased lateral root length at 75 mM NaCl (CSHL #4 D4a **Fig. S21-23**). Two out of two independent transformants for RNAi line #16 showed a decreased relative main root length and increased lateral root number at 75 mM NaCl (CSHL #16 A2c and B1a, **Fig. S21-23**). In three of the four screened independent transformants of line #19 we observed an increased lateral root length at 75 mM NaCl (CSHL #19 F1a, I3c, and L1c **Fig. S21-23**). We did not observe strong reduction in lateral root development across RNAi lines targeting two closely related DUF247 genes (i.e., RNAi





**Figure 6.**

***sr3g* mutant displays reduced growth rate for the lateral root length and Na<sup>+</sup> content in the root.**

Root system architecture analysis of Col-0 and *sr3g-5* genotypes under various concentrations of NaCl are shown here. **(A)** Shown are representative images of 13-day-old Col-0 and *sr3g-5* genotypes that experienced 9 days of salt treatment at indicated concentrations as well as growth rate for lateral root length. **(B)** Salt Tolerance Index (STI) for the main root length, average lateral root length, and lateral root number at 75 mM NaCl. The STI was calculated by dividing the growth rate measured under salt stress by the growth rate measured under control condition for each genotype. **(C)** Na<sup>+</sup>/K<sup>+</sup> ratio in root and shoot of Col-0 and *sr3g-5* after two weeks on 75 mM salt are shown. Each dot in (A) and (B) represents individual replicate per genotype. Lines in (A) and (B) graphs represent median and average, respectively. In (A) and (B), the asterisks above the graphs indicate significant differences between Col-0 and the *sr3g-5* mutant, as determined by the Student's t-test: \**P* < 0.05. Statistical analysis in (C) was done by comparison of the means for all pairs using Tukey-Kramer HSD test. Levels not connected by the same letter are significantly different (*P* < 0.05). Root system architecture and ICP-AES analyses of Col-0 and *sr3g-5* mutant are shown in details in **Figure S17**.

line #22, targeting AT3G50150 and *SR3G*), as in the single mutant (**Fig. 6**, **Fig. S21**). The above results further support that these two genes do not act redundantly in the context of lateral root development.

## Loss of *SR3G* results in increased salt tolerance in soil

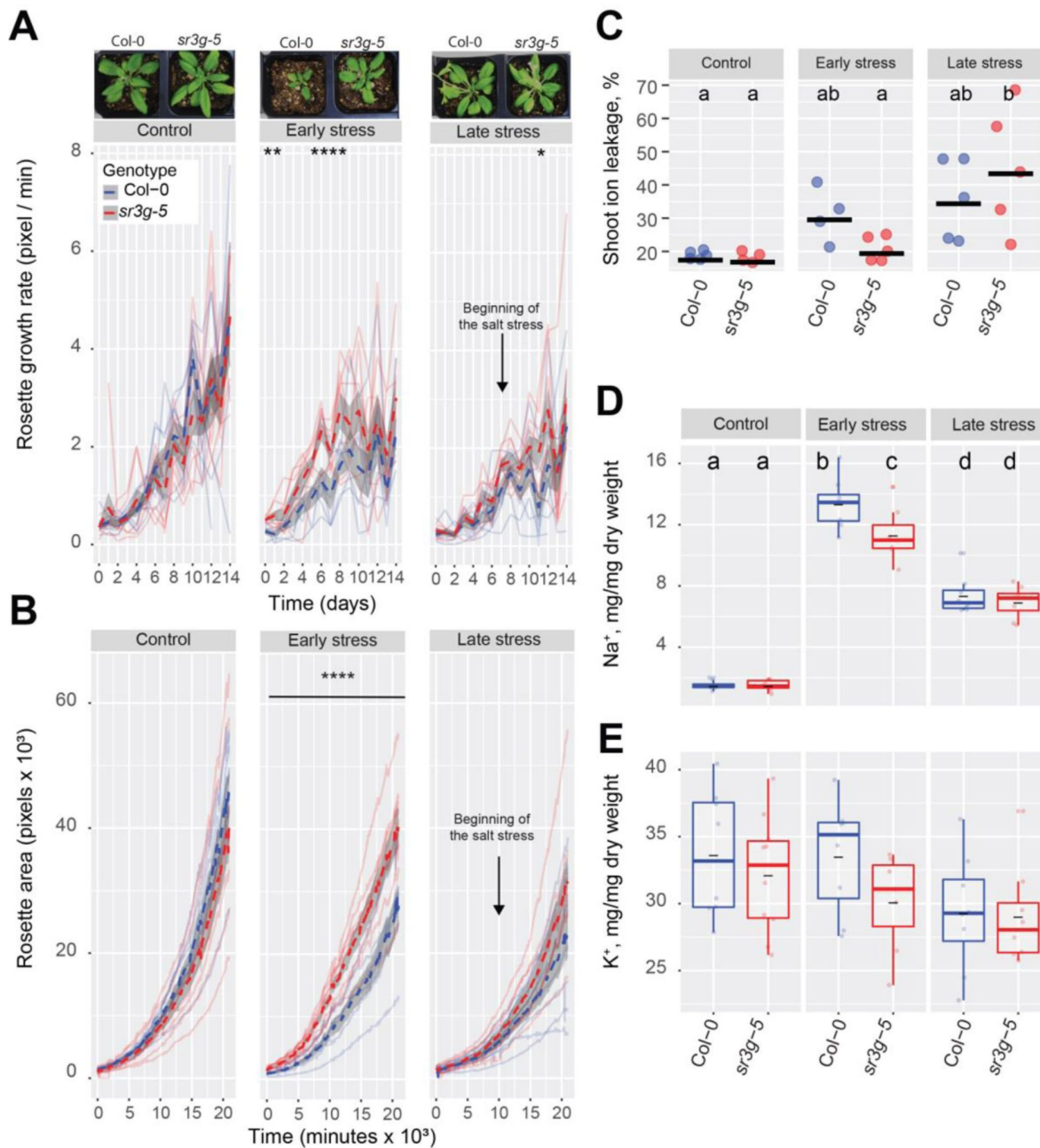
To investigate *SR3G*'s contribution to salt stress tolerance beyond seedling stage and agar plate conditions, we evaluated the performance of Col-0 and *sr3g-5* mutant in soil. We exposed the soil-grown plants to salt stress and imaged the rosette size every 30 minutes for two weeks using the automated imaging platform (L. 'ang Yu et al. 2023). As salt-induced changes depend on the developmental stage at which the stress is applied (Julkowska et al. 2017), we examined the role of *SR3G* exposed to salt stress at 2 or 3 weeks after germination, corresponding to early and late vegetative stages. We calculated a daily growth rate for each plant, using the projected rosette area. No significant difference was observed between Col-0 and *sr3g-5* mutant under control conditions (**Fig. 7A-B**), indicating that *SR3G* does not play a role in plant development under non-stress condition. Significant difference was observed between Col-0 and *sr3g-5* for plants exposed to early salt treatment (**Fig. 7A-B**), where *sr3g-5* mutant displayed an increased rosette growth rate as compared to Col-0 (**Fig. 7A**). Late stress exposure resulted in significantly higher *sr3g-5* growth rate at two and three days after exposure to salt stress (**Fig. 6A**). Similarly, the *sr3g-5* mutant showed a significantly higher rosette area compared to Col-0 exclusively when plants were exposed to salt stress at an earlier stage of their development (**Fig. 7A**).

To investigate the role of *SR3G* in the maintenance of cellular integrity during salt stress, we evaluated the salt-induced changes in cell membrane integrity by measuring ion leakage in the leaf discs at the end of the experiment (**Fig. 7C**). While ion leakage increased significantly in Col-0 under early salt stress conditions, no salt-induced change was observed for *sr3g-5* compared to control condition (**Fig. 7C**). Although the *sr3g-5* mutant exhibited increased ion leakage in response to late salt stress, this phenomenon may be more closely associated with the onset of senescence rather than direct exposure to salt stress (**Fig. 7C**).

We further explored the role of *SR3G* in ion exclusion by quantifying the accumulation of Na<sup>+</sup> and K<sup>+</sup> ions in the rosette at the end of the experiment (4-weeks-old plants). We observed a reduced accumulation of Na<sup>+</sup> in *sr3g-5* mutants compared to Col-0 for plants exposed to salt stress at an earlier timepoint (**Fig. 7D**). No significant differences in Na<sup>+</sup> accumulation was observed between the Col-0 and *sr3g-5* at the late salt stress regime (**Fig. 7D**). A subtle, non-significant, reduction in K<sup>+</sup> content was observed in *sr3g-5* mutant compared to Col-0 across all three conditions studied (**Fig. 7E**). In summary, these results suggest that *SR3G* plays a negative role in maintenance of plant growth, cellular integrity and Na<sup>+</sup> accumulation during the salt stress, especially when the salt stress is applied during early vegetative development.

## *sr3g* mutant displays enhanced root suberization

As *SR3G* limits shoot growth, promotes cell damage and sodium translocation into shoot (**Fig. 7**), and is expressed in the most inner layers of the root (i.e., stele, **Fig. S13**), we hypothesized that *SR3G* plays a role in the regulation of suberization within the Casparian strips located in the endodermis, the layer adjacent to the stele. Salt stress exposure increases root suberization, thereby limiting Na<sup>+</sup> loading into the transpiration stream and accumulation in the shoot (Baxter et al. 2009). We examined root suberization patterns in Col-0 and *sr3g-5* under non-stress and salt stress conditions. Root suberization was visualized using lipophilic stain Fluorol Yellow (FY), and staining intensity was examined under a confocal microscope. Additionally, we examined the developmental transition in root suberization through imaging four regions along the main root (**Fig. 8**). Visually, we observed that roots of *sr3g-5* were more suberized compared to Col-0 under both control and salt stress conditions (**Fig. 8A**). The quantification of FY signal intensity revealed higher levels of root suberization in *sr3g-5* compared to Col-0 in sections one to three of the root under control condition (**Fig. 8B**). Only in root section four, the level of suberization



**Figure 7.**

***sr3g* mutant has higher growth rate and larger rosette area while displaying less ion leakage and Na<sup>+</sup> accumulation in shoot under salt stress.**

The salt stress responses of 2-week-old (referred to as “early salt stress”) or 3-week-old (referred to as “late salt stress”) soil-grown Col-0 and *sr3g-5* plants that were exposed to a final concentration of 100 mM NaCl were examined here. **(A)** Rosette growth rate and **(B)** area were monitored over a period of two weeks, during which the “early salt stress” group experienced two weeks of salt stress while the “late salt stress” group experienced one week. Representative images of 4-week-old Col-0 and *sr3g-5* genotypes were shown in (A). **(C)** Ion leakage percentage, **(D)** Na<sup>+</sup> and **(E)** K<sup>+</sup> contents in shoot were measured in 4-week-old plants. Each line in (A) and (B) represents the trajectory per genotype through time, where red and blue lines indicate Col-0 and *sr3g-5* plants grown under control or salt-stress conditions, respectively. The dashed lines represent the mean values of the genotype per condition, whereas the grey band represents the confidence interval. The asterisks above the graphs indicate significant differences between Col-0 and *sr3g-5* plants, as determined by one-way analysis of variance: \**P* < 0.05, \*\**P* < 0.01, \*\*\**P* < 0.001, and \*\*\*\**P* < 0.0001. Statistical analysis in (C) to (E) was done by comparison of the means for all pairs using Tukey-Kramer HSD test. Levels not connected by the same letter are significantly different (*P* < 0.05).

significantly decreased in *sr3g-5* compared to Col-0 (**Fig. 8B**). In plants exposed to salt stress, we observed a non-significant difference between *sr3g-5* and Col-0 throughout the four sections, although, in section one which is the root tip proximal, *sr3g-5* showed more suberization compared to Col-0 (section 1, **Fig. 8A**). These results suggest that *SR3G* may play a negative role in root suberization.

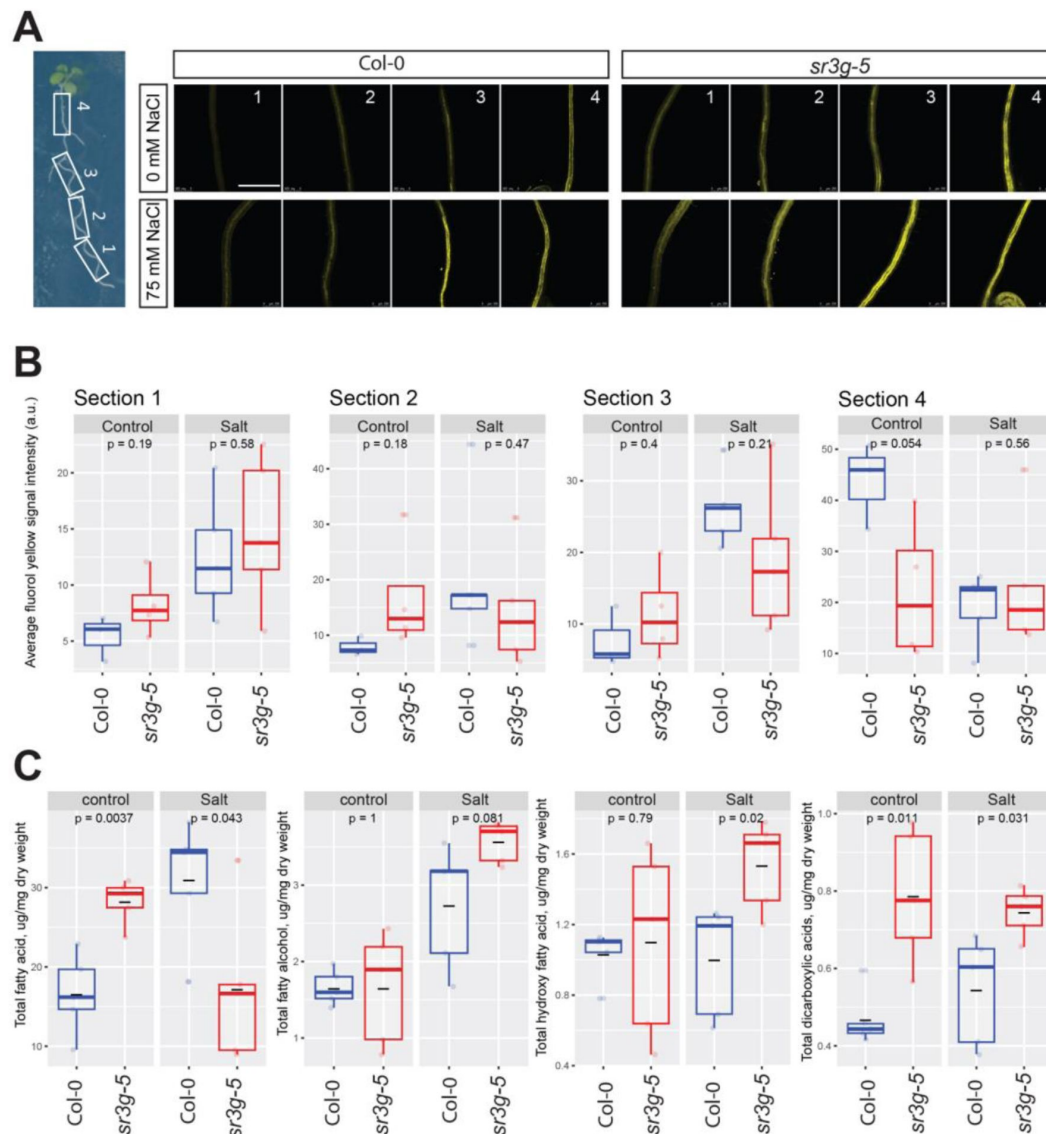
To validate the FY histochemical analysis, we quantified root suberin monomers in the *Arabidopsis* seedlings using GC-MS analysis. Suberin monomers that were detected in the *Arabidopsis* roots included fatty acid (FA), fatty alcohol (OH-FA),  $\alpha,\omega$ -dicarboxylic acids (DCA), and  $\omega$ -hydroxy fatty acid ( $\omega$ OH-FA). Roots of *sr3g-5* contained significantly more total FA and DCA monomers compared to Col-0 under control condition, whereas the total OH-FA and  $\omega$ OH-FA remained unchanged (**Fig. 8C**). In plants exposed to salt stress, the *sr3g-5* mutant showed an increased abundance of three suberin monomers, with DCA and  $\omega$ OH-FA showing the most significant changes (**Fig. 8C**). The total FA decreased significantly in the mutant compared to Col-0 (**Fig. 8C**). Together, these results further corroborate with FY staining, suggesting that *SR3G* plays a negative role in root suberization.

### ***SR3G* expression is regulated by *WRKY75* transcription factor**

To identify potential transcriptional regulators of *SR3G*, we examined the *SR3G* promoter for cis-regulatory elements using the CisCross tool (<https://plamorph.sysbio.ru/ciscross/>), and made a list of transcription factors that putatively interact with *SR3G* promoter (**Fig. 9A**). As the root stele showed the highest enrichment in the *SR3G* transcript (**Fig. S13**), we examined the transcription factors that were previously identified to be expressed in root stele (Brady et al. 2011). Cross-referencing of these two datasets revealed one transcription factor, *WRKY75* (AT5G13080) that was both enriched in root stele, and potentially binding to *SR3G* promoter.

*WRKY75* was previously described to act as a positive regulator of salt stress response by enhancing *SOS1* expression (Lu et al. 2023). To investigate whether the *SR3G* expression depends on *WRKY75*, we examined the *SR3G* expression in the *wrky75* T-DNA insertion lines. *SR3G* expression was undetected in *wrky75* loss-of-function mutants (**Fig. 9B**), suggesting that *WRKY75* regulates *SR3G* expression. In comparison, expression of *WRKY75* remained unchanged in the *sr3g-5* mutant when compared to Col-0 (**Fig. 9C**). To investigate whether *WRKY75* directly binds to the *SR3G* promoter, we performed Y1H assay. Similar to the two positive controls, the strains transformed with *WRKY75*-AD and *SR3G*:pLacZ grew on the SD/-Trp/-Ura + X-Gal medium, thereby confirming the interaction between *WRKY75* and *SR3G* promoter region (**Fig. 9D**). In contrast, *WRKY75* did not show binding affinity to the promoter region of the adjacent gene to the *SR3G* (i.e., DUF247-150, AT3g50150), thereby reinforcing the uniqueness of the interaction between the *SR3G* and *WRKY75* (**Fig. 9E**).

To further examine the functional consequences of the relationship between *SR3G* and *WRKY75*, we generated *wrky75-3/sr3g-5* double mutants, and examined their root system architecture alongside the single mutants and Col-0. Under control conditions, both *sr3g-5* and *wrky75-3* mutants showed significant reduction in main root length compared to Col-0 and the two double mutants (**Fig. 10A**). Under 75 mM salt, only *wrky75-3* showed significant reduction in main root length compared to all other genotypes tested, confirming the previously described salt sensitivity for *wrky75* mutant under salt stress (Lu et al. 2023). We have not observed any differences in main root length under 125 mM salt across all genotypes tested (**Fig. 10A**). Regarding lateral root number (**Fig. 10B**) and length (**Fig. 10C**), we observed significant reduction in lateral root number and length for the two single mutants, *sr3g-5* and *wrky75-3*, compared to other genotypes only under control condition (**Fig. 10B-C**). We further investigated  $\text{Na}^+$  and  $\text{K}^+$  accumulations as well as  $\text{Na}^+/\text{K}^+$  ratio in plate grown seedlings (**Fig. 10D**, **Fig. S24**). At 75 mM salt concentration, a subtle reduction in root  $\text{Na}^+/\text{K}^+$  ratio was observed in the *wrky75* single mutant compared to the other genotypes tested. Conversely, shoot  $\text{Na}^+/\text{K}^+$  ratio exhibited a slight decrease in the two double mutants compared to the other three genotypes at 75 mM salt concentration. At 125 mM



**Figure 8.**

### *sr3g-5* mutant root is more suberized than Col-0.

**(A)** Representative images of fluorol yellow (FY) staining of four root sections (# 1 to 4, from root tip to root base as shown by white rectangles across the seedling) were shown for Col-0 and *sr3g-5* mutant with or without 75 mM NaCl. Seeds were germinated on the 1/2 MS plates for 4 d and then seedlings were transferred to the treatment plates with or without 75 mM NaCl for 2 more days. **(B)** Quantification of FY signal intensity for each root section. The FY quantification was done using ImageJ. Scale bar is equal to 500  $\mu$ m in all. **(C)** Suberin monomers detected using Gas Chromatography–Mass Spectrometry (GC-MS) in the Col-0 and *sr3g-5* mutant roots with or without 75 mM NaCl included fatty acid (FA), fatty alcohol (OH-FA),  $\alpha,\omega$ -dicarboxylic acids (DCA), and  $\omega$ -hydroxy fatty acid ( $\omega$ OH-FA). The significant differences between Col-0 and *sr3g-5* mutant were evaluated using Student's t-test.





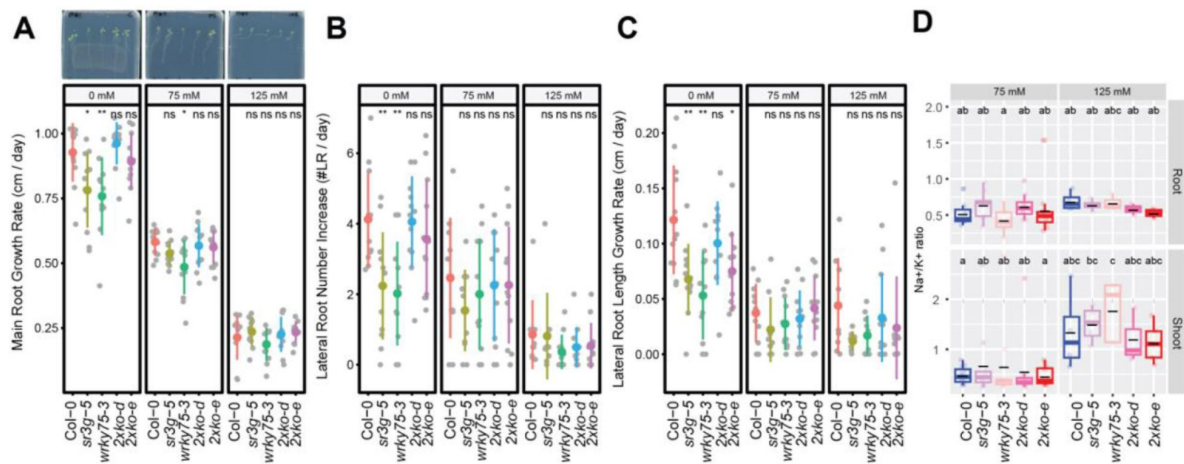
salt concentration, root  $\text{Na}^+/\text{K}^+$  ratio decreased in the two double mutants compared to Col-0 and the two single mutants. Similarly, the two double mutants maintained lower  $\text{Na}^+/\text{K}^+$  ratios compared to Col-0 and the two single mutants in shoot at 125 mM salt concentration (**Fig. 10D**), indicating that the *SR3G* mutation indeed compensated for the increased  $\text{Na}^+$  accumulation observed in the *wrky75* mutant under salt stress.

## Discussion

Exposure to abiotic stress alters plant architecture by affecting biomass distribution between individual organs (Julkowska and Testerink 2015; Guo et al. 2020). Exposure to drought alters the distribution of the biomass between root and shoot, to optimize water uptake, while limiting transpiration (Asch et al. 2005; Karcher et al. 2008; Uga et al. 2013; Chen et al. 2022). While the processes underlying salt-induced changes in shoot and root growth dynamics and architecture have been systematically described (Geng et al. 2013; Duan et al. 2013; Julkowska et al. 2017; Deolu-Ajayi et al. 2019; Yutao Zou et al. 2022; Awlia et al. 2021; Ariel et al. 2010), the methodological exploration of biomass distribution between root and shoot for plants exposed to salt stress is underexplored in the current body of research. Here, we developed a tool that allows a non-destructive quantification of root and shoot projected area of the seedlings grown on agar plates (**Fig. 1A**). This new tool allows us to efficiently describe changes in biomass accumulation and stress-induced alterations in root and shoot growth. Although the method is restricted to small plants and early developmental stages, it provides valuable insight into stress responses. The application of this tool to an *Arabidopsis* natural diversity panel revealed that salt stress exposure results in loss of coordination between root and shoot growth (**Fig. 1D**), rather than a clear directional change, as in the case of drought. To our knowledge, this is the first systematic study of changes in root:shoot ratio. The developed tool for quantifying root and shoot size in plate-grown *Arabidopsis* has the potential to significantly advance research by enabling developmental and stress studies, focusing on coordination of root and shoot development during seedling establishment. Facilitating detailed comparisons of root:shoot ratios, this tool can aid in formulating new hypotheses about the mechanisms contributing to seedling vigor and stress tolerance.

The natural variation in salt-induced changes in root:shoot ratio revealed a clear association with SNPs on chromosome 3 within *SR3G*, that is situated within a region containing eight other *DUF247* genes (**Fig. 2**). *SR3G* was previously associated with self-incompatibility in grasses (Lian et al. 2021; Crain et al. 2020), as well as with lesions and dwarfism in *Arabidopsis* (Kondou et al. 2013) through constitutive activation of defense responses. However, the *SR3G* associated with constitutive defense response was AT3G60470 (Kondou et al. 2013), which is located outside the interval associated with salt stress-induced changes in root:shoot ratio (**Fig. 2C**). Because we did not observe any consistent phenotypes in the T-DNA insertion lines of *DUF247*s neighboring the *SR3G* (**Fig. S19-20**), we examined the RNAi lines targeting multiple *DUF247*s including *SR3G* (**Fig. S21-23**). Our root system architecture analyses revealed no redundancy between *SR3G* and other *DUF247*s, and the reduced lateral root development in double mutants of RNAi lines simply mimics the phenotype of *sr3g-5* single mutant (**Fig. 6**, **Fig. S21**). Further, we demonstrated that only *SR3G* is regulated by *WRKY75* (**Fig. 9B-E**). As these genes are result of tandem duplication, they are expected to accumulate a lot of variation in the cis-regulatory regions. Therefore, differential regulation of two closely related genes can be expected by the accumulated variation within the promoter region.

The results acquired in this study provide multiple layers of evidence that support the role of *SR3G* as a negative regulator of salt stress response in *Arabidopsis*. Loss-of-function *SR3G* maintained higher shoot growth rate and larger rosette area when exposed to salt stress in the soil (**Fig. 7A-B**), probably due to reduced  $\text{Na}^+$  accumulation (**Fig. 7D**), which in turn resulted in lower rates of cellular damage (**Fig. 7C**). This increased salt tolerance feature can be explained by the



**Figure 10.**

***wrky75/sr3g* double mutants roots exhibit reduced sensitivity to salt stress and accumulate lower levels of  $\text{Na}^+$  in their shoots.**

(A-C) Root system architecture analysis for the *Col-0*, *sr3g-5*, *wrky75-3*, and two double mutants under various concentrations of NaCl are shown here. Main root growth rate (A), Lateral root number increase (B), and Lateral root length increase (C) are shown for the indicated genotypes at various salt concentrations. (D)  $\text{Na}^+/\text{K}^+$  ratio in root and shoot of indicated genotypes after two weeks on treatment plates. Each dot represents individual replicate per genotype. The asterisks above the graphs (A-C) indicate significant differences between *Col-0* and the genotype, as determined by the Student's t-test: \* $P < 0.05$  and \*\* $P < 0.01$ . ns denotes no statistically significant. Statistical analysis in (D) was done by comparison of the means for all pairs using Tukey-Kramer HSD test. Levels not connected by the same letter are significantly different ( $P < 0.05$ ).

increased root suberization (**Fig. 7**) as suberin-deficient mutants show susceptibility to salt stress (Barberon et al. 2016; Ursache et al. 2021; Shukla et al. 2021). For example, *Arabidopsis CYP86A1/HORST* (At5g58860) encodes a cytochrome P450-dependent enzyme that is responsible for catalyzing fatty acid  $\omega$ -hydroxylation. It plays a pivotal role in the biosynthesis of aliphatic suberin, particularly expressed in the root endodermis. Knock-out mutations of *CYP86A1/HORST* result in a substantial decrease of approximately 60% in total root suberin (Y. Li et al. 2007; Höfer et al. 2008). Consequently, the diminished aliphatic suberin content in the roots of *cyp86a1* mutants leads to increased permeability to water and solutes (Ranathunge and Schreiber 2011). Interestingly, *SR3G* expression is enhanced by salt stress (**Figs. 4**, **Fig. S13**), specifically in the root stele (**Fig. S13B**). The root stele encompasses vascular bundles, procambium and pericycle, which gives rise to the secondary growth and lateral root primordia respectively (Birnbaum et al. 2003). The stele is central to the systemic transport of water, minerals, and nutrients within the plant (B. Li et al. 2017; Ramachandran et al. 2020; Zarebanadkouki et al. 2019), and to the exclusion of salt ions from the transpiration stream (Møller et al. 2009).

The deposition of hydrophobic residues of lignin and suberin in the Casparian strips poses a significant barrier to apoplastic transport (Foster and Miklavcic 2017). Previously, silencing genes involved in Casparian strips formation resulted in increased lateral root development (X. Wang et al. 2020). As the suberization of the Casparian strips increases upon salt stress, it limits the accumulation of  $\text{Na}^+$  and loss of  $\text{K}^+$  ions from the root stele (Barberon et al. 2016). However, the effects of increased root suberization under salt stress have thus far not been described to affect lateral root development. Here, we have identified *SR3G* as a negative regulator of root suberization under salt stress (**Fig. 8**), which provides an additional layer of protection against sodium accumulation (**Fig. 8**), but negatively impacts lateral root development (**Fig. 6A-B**).

As *SR3G* was previously not described in the context of salt stress, we attempted to identify its molecular context through its transcriptional regulation. Through our *in-silico* analysis, we have identified *WRKY75* as a putative regulator of *SR3G* expression, and confirmed it using RT-qPCR and Y1H assay (**Fig. 9**). *WRKY75* was previously described to act as a positive regulator of salt stress tolerance through activation of *SOS1* expression (Lu et al. 2023). Similar to previous study, we also observed *wrky75* mutants to be more sensitive to salt stress within our experiments (**Fig. 10A-C**). Nevertheless, when the *wrky75* mutation was coupled with *sr3g-5*, the roots displayed a phenotype identical to Col-0 (**Fig. 10A**), implying that the salt sensitivity of the *wrky75* mutant was counteracted by the *SR3G* mutation. Considering *WRKY75*'s role in positively regulating *SOS1* expression during salt stress, it is plausible that its binding to the *SR3G* promoter inhibits its expression, thereby enhancing salt tolerance on a larger scale.

One of the main players in  $\text{Na}^+$  transport is High-affinity  $\text{K}^+$  Transporter (*HKT1*) (Laurie et al. 2002). *HKT1* is expressed in root stele (Mäser et al. 2002), where it retrieves  $\text{Na}^+$  from the transpiration stream and limits ion accumulation in the shoot, resulting in higher salt stress tolerance at later developmental stages (Møller et al. 2009). At early developmental stages, *HKT1* overexpression reduces the development of lateral root primordia (Julkowska et al. 2017), thus creating less  $\text{Na}^+$  entry gate into the main root, resulting in less  $\text{Na}^+$  accumulation in shoot. Similarly, *sr3g-5* mutants developed fewer lateral roots that are consistent with the lower  $\text{Na}^+$  levels in the root upon salt stress (**Fig. 6**). The effects of *HKT1* and *SR3G* differ in the magnitude of repression of lateral root development, with *HKT1* mutants resulting in almost complete cessation of lateral root development (Julkowska et al. 2017), whereas repression of *SR3G* function has much more subtle phenotype. Whereas the effect of cell-type specific overexpression of *HKT1* on plant performance was negative during the early developmental stage, due to impairing lateral root development, the *sr3g* mutants showed increased salt tolerance when exposed to salt stress treatment during their early vegetative growth (**Fig. 7**). These results suggest that reduced lateral root development under salt stress can have a gradient of responses

leading to increased and decreased salt tolerance, and the effect is highly dependent on the wider physiological mechanisms, and their contributions to ion exclusion and maintained plant development throughout salt stress exposure.

In conclusion, the study of salt-induced changes in the root:shoot ratio, alongside the development of new tools to quantify these changes during the seedling establishment stage, represents an important new perspective into mechanisms underlying salt stress tolerance. Identification of SR3G as a novel regulator of root suberization, shoot growth, and Na<sup>+</sup> accumulation provides better understanding of molecular mechanisms underlying plant responses to salt stress, but also offers a new avenue to explore more nuanced and tissue-specific strategies for enhancing salt stress tolerance.

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## Author contributions

M.M.J. and M.R.I. conceived the project and wrote the manuscript with inputs from all coauthors, C.T. and M.A.T. helped with conceiving the overall idea for the project, infrastructure and funding, R.A.H. and R.S. developed the tool to quantify root and shoot size from Arabidopsis grown on agar plates, A.K. assisted with GWAS analysis, M.R.I. conducted majority of the experiments for functional characterization of SR3G, H.S., and M.D.A. performed molecular cloning of DR3G, H.S. screened other DUF247s T-DNA and RNAi mutant lines, L.J. performed the analysis of DUF247 domains that informed the cloning of individual SR3G fragments, G.D. and M.W. conducted GC-MS on suberin monomers, M.P. and E.C. performed ICP-AES, A.D.L.N. and L.Y. performed phylogenetic analysis, B.L. and Y.H. conducted Y1H.

## Competing interests

The authors declare no competing interests.

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## Reviewer #1 (Public Review):

Summary:

The authors aim to assess the effect of salt stress on root:shoot ratio, identify the underlying genetic mechanisms, and evaluate their contribution to salt tolerance. To this end, the authors systematically quantified natural variations in salt-induced changes in root:shoot ratio. This innovative approach considers the coordination of root and shoot growth rather than exploring biomass and the development of each organ separately. Using this approach, the authors identified a gene cluster encoding eight paralog genes with a domain-of-unknown-function 247 (DUF247), with the majority of SNPs clustering into SR3G (At3g50160). In the manuscript, the authors utilized an integrative approach that includes genomic, genetic, evolutionary, histological, and physiological assays to functionally assess the contribution of their genes of interest to salt tolerance and root development.

Strengths:

The holistic approach and integrative methodologies presented in the manuscript are essential for gaining a mechanistic understanding of a complex trait such as salt tolerance. The authors focused on At3g50160 but included in their analyses additional DUF247 paralogs, which further contributes to the strength of their approach. In addition, the authors considered the developmental stage (young seedlings, early or late vegetative stages) and growth conditions of the plants (agar plates or soil) when investigating the role of SR3G in salt tolerance and root or shoot development.

Weaknesses:

The authors' claims and interpretation of the results are not fully supported by the data and analyses. In several cases, the authors report differences that are not statistically significant (e.g., Figures 4A, 7C, 8B, S14, S16B, S17C), use inappropriate statistical tests (e.g., t-test instead of Dunnett Test/ANOVA as in Figures 10B-C, S19-23), present standard errors that do not seem to be consistent with the post-hoc Tukey HSD Test (e.g., Figures 4, 9B-C, S16B), or lack controls (e.g., Figure 5C-E, staining of the truncated versions with FM4-64 is missing).

In other cases, traits of root system architecture and expression patterns are inconsistent between different assays despite similar growth conditions (e.g., Figures S17A-B vs. 10A-C vs. 6A, and Figures S16B vs. 4A/9B), or T-DNA insertion alleles of WRKY75 that are claimed to be loss-of-function show comparable expression of WRKY75 as WT plants. Additionally, several supplemental figures are mislabeled (Figures S6-9), and some figure panels are missing (e.g., Figures S16C and S17E).

Consequently, the authors' decisions regarding subsequent functional assays, as well as major conclusions about gene function, including SR3G function in root system architecture, involvement in root suberization, and regulation of cellular damage are incomplete.

<https://doi.org/10.7554/eLife.98896.1.sa1>

**Reviewer #2 (Public Review):**

Salt stress is a significant and growing concern for agriculture in some parts of the world. While the effects of sodium excess have been studied in *Arabidopsis* and (many) crop species, most studies have focused on Na uptake, toxicity, and overall effects on yield, rather than on developmental responses to excess Na, per se. The work by Ishka and colleagues aims to fill this gap.

Working from an existing dataset that exposed a diverse panel of *A. thaliana* accessions to control, moderate, and severe salt stress, the authors identify candidate loci associated with altering the root:shoot ratio under salt stress. Following a series of molecular assays, they characterize a DUF247 protein which they dub SR3G, which appears to be a negative regulator of root growth under salt stress.

Overall, this is a well-executed study that demonstrates the functional role played by a single gene in plant response to salt stress in *Arabidopsis*.

The abstract and beginning of the Discussion section highlight the "new tool" developed here for measuring biomass accumulation. I feel that this distracts from the central aims of the study, which is really about the role of a specific gene in root development under salt stress. I would suggest moving the tool description to less prominent parts of the manuscript.

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